

Experiment 5.1: Design and Analysis of an FIR High-Pass Filter Using the Rectangular Window Technique in MATLAB

Aim: To design and analyze an FIR High-Pass Filter using the Rectangular Window technique in MATLAB.

MATLAB Code

```
clc;
clear;
close all;

%% Save Command Window Output
diary('Exp5_1_Output.txt');

%% --- Input Parameters ---
fp = 1000;      % High-pass cutoff frequency (Hz)
fs = 8000;      % Sampling frequency (Hz)
N = 50;         % Filter order

wc = fp/(fs/2); % Normalized cutoff (0-1)

%% --- FIR High Pass using Rectangular Window ---
b_rect = fir1(N, wc, 'high', rectwin(N+1));

disp('--- High-Pass Coefficients (Rectangular Window) ---');
disp(b_rect);

%% --- Frequency Response Plot ---
figure(1);
freqz(b_rect,1);
title('High-pass FIR using Rectangular Window');

%% Stop Saving Command Window
diary off;

%% Save Figure (300 DPI)
exportgraphics(figure(1), 'Exp5_1_Rectangular_Window.png', 'Resolution', 300);

%% Completion Message
disp(' ');
disp('=====');
disp('Experiment 5.1 Completed Successfully');
disp('Files Saved:');
disp('Exp5_1_Output.txt');
disp('Exp5_1_Rectangular_Window.png');
disp('=====');
```

Output

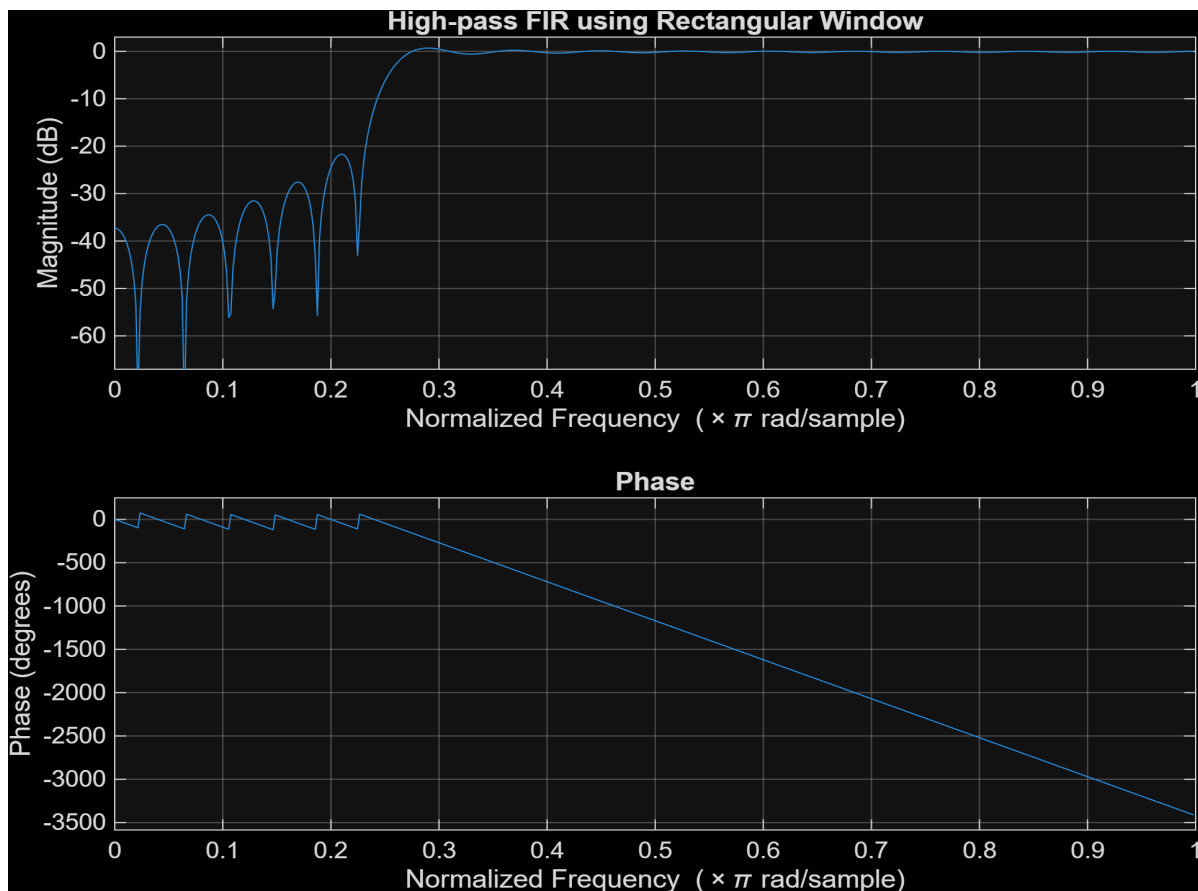
```
--- High-Pass Coefficients (Rectangular Window) ---
-0.0089
  0
  0.0097
  0.0143
  0.0106
  0
-0.0117
-0.0175
-0.0131
  0
  0.0148
  0.0225
  0.0171
  0
-0.0202
-0.0314
-0.0247
```

```

0
0.0318
0.0524
0.0445
0
-0.0741
-0.1572
-0.2223
0.7407
-0.2223
-0.1572
-0.0741
0
0.0445
0.0524
0.0318
0
-0.0247
-0.0314
-0.0202
0
0.0171
0.0225
0.0148
0
-0.0131
-0.0175
-0.0117
0
0.0106
0.0143
0.0097
0
-0.0089

```

Output Image



Result: The FIR High-Pass filter using the Rectangular Window technique was successfully designed and analyzed in MATLAB. The frequency and phase responses were obtained successfully.